NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

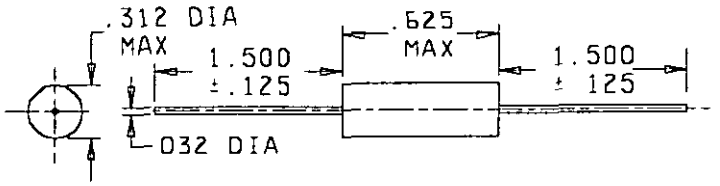
MIL-PRF-27/88
22 April 1976

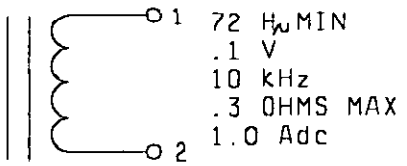
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE),
INDUCTOR, POWER, TYPE TF5S04ZZ905

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the inductor described herein shall consist of this document and the latest issue of Specification MIL-T-27.





INCHES	MM
.032	0.81
.125	3.18
.312	7.92
.625	15.88
1.500	38.10

WV: 50 VOLTS MAX
ALTITUDE: 70,000 FT MAX

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Marking shall be on the side of the case.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the extreme terminals of the winding.)

Electrical ratings.
Inductance (1-2) 72 µH minimum.
Voltage (1-2): 0.1 Vrms, 10 kHz.
Power. 0.3 watt.
Current (1-2) 1 ampere dc.
DC resistance (1-2) 0.3 ohm dc maximum.
Working voltage: 50 V maximum.

Design and construction
Dimensions and configuration See figure
Duty cycle: Continuous.
Case Encapsulated
Material. Epoxy
Altitude 70,000 feet maximum.
Terminals Solid wire lead type, tinned soft copper.
Weight 3.2 grams
Operating temperature range -55° to 130°C

Terminal strength MIL-STD-202, method 211, condition A, 2 pounds.

Dielectric withstanding voltage
At sea level 100 Vrms
At barometric pressure 50 Vrms

Electrical characteristics
Inductance With 0.1 Vrms, 10 kHz across, and 1 ampere dc applied to (1-2) the inductance shall be 72 µH minimum
DC resistance at 25°C 0.3 ohm across (1-2) maximum
Self resonant frequency 2.5 MHz minimum

Temperature rise 30°C maximum, with 0.1 Vrms, 10 kHz, 1 ampere dc applied to (1-2) at an ambient temperature of 100°C

Marking location See figure

Part number M27/88-01

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 80
DSA - ES

User activities
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity
Army - EL

Agent
DSA - ES

(Project 5950-0506-8)

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R25S

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27/88 INDUCTOR, POWER, TYPE TF5S04ZZ905

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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